

required for a gateway. It was also decided to use HTTP and WebSockets for the TCP/IP protocols with HTTP requests and responses used for all operations except the delivery of streams of Bluetooth characteristic notifications (which were delivered over a WebSocket).

For HTTP support, the Apache Web Server was chosen as it has a wide collection of modules and the open source websocketd was used for web socket support. A webcam was also plugged into the gateway to observe remote actions.

The adapter code was written

in Python. To integrate the Python script with the web server in HTTP, the common gateway interface (CGI) was used.

BlueZ is the name of the Bluetooth stack you will find on Linux. It provides a language-agnostic API, which uses the Linux d-bus service that allows inter-process communication to take place.

Bluetooth Internet gateway implementation

HTTP GET. Requests which do not change the state of remote objects with an HTTP query string used to pass associated parameters.

HTTP PUT. Requests which do change the state of remote objects with JSON objects in the HTTP request body used to encode and transport associated parameters.

RESPONSES. HTTP status and JSON object.

Here's an example that concerns device discovery (Fig. 7):

The above shows a response generated by an adapter script called `do_discover_devices.py`. It's invoked using HTTP GET and the scan time parameter is passed into the query string. This indicates how long the gateway should spend performing Bluetooth scanning in milliseconds. The response is an array of objects representing the devices discovered in JSON format.

In this example, HTTP PUT was used (Fig. 8) for writing to a Bluetooth characteristic. Here the Bluetooth device address is assigned as the first parameter and identifies the device to interact with.

The handle parameter is a unique identifier for a specific instance of a Bluetooth characteristic, which belongs to a particular Bluetooth service.

Bluetooth Internet gateway security

Anything that allows access to your device over the Internet needs to be secure. A review of the security (Fig. 11) of the initial gateway design and implementation identified numerous security issues.

For example, there was no authentication mechanism and therefore no control over who may or may not access the gateway.

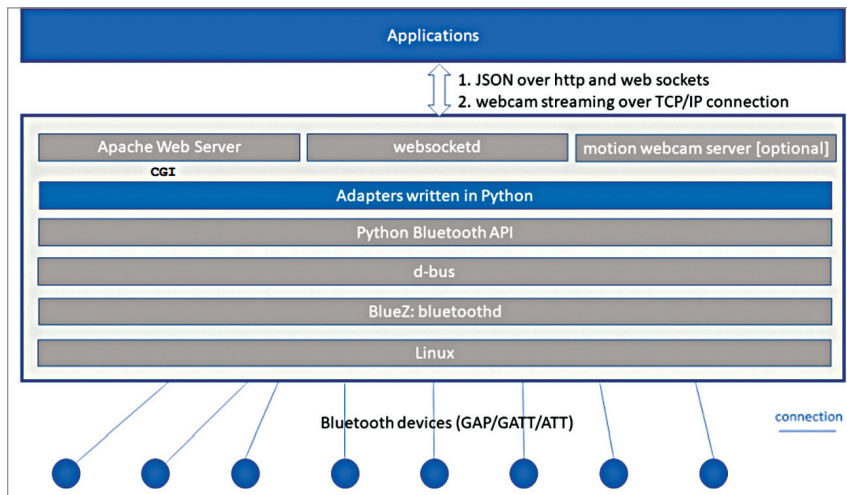


Fig. 6: Gateway physical architecture

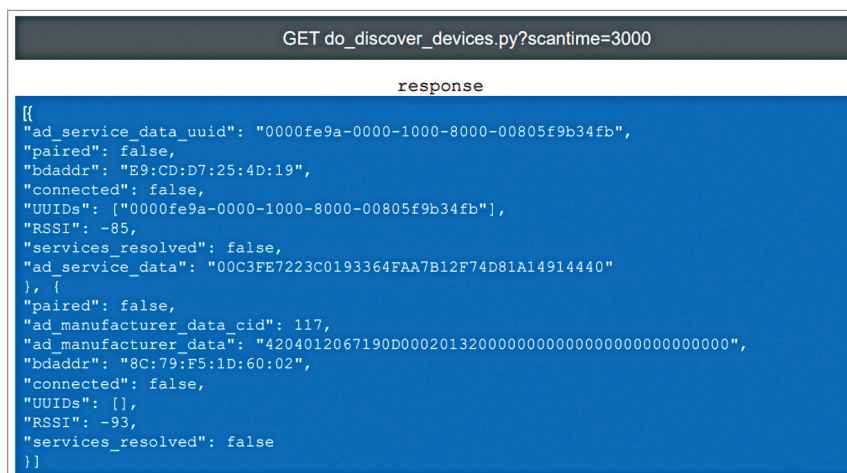


Fig. 7: Device discovery

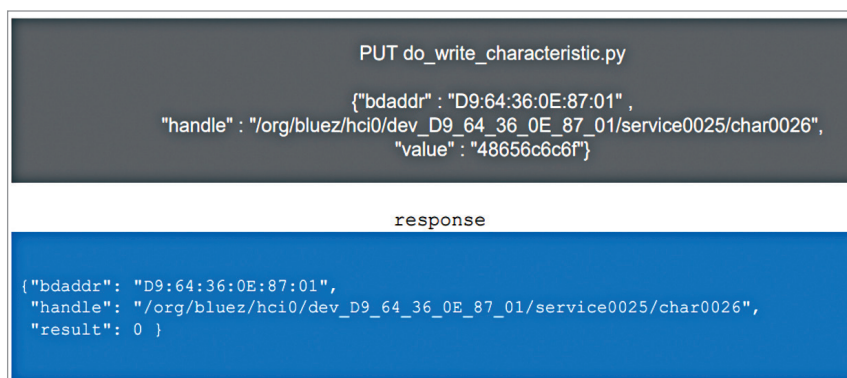


Fig. 8: Writing to a Bluetooth characteristic